

Generation Interconnection System Impact Study Report

For Queue Project Q19-1318

Michigan Wind 2 138 kV

January 30, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1318. The Interconnection Customer proposes a 153.4 MW solar photovoltaic facility interconnecting at the Michigan Wind 2 138 kV substation in the Virginia area. The request was received on October 26, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 458 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,441,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gainesville / Novec Paradise 138 KV	Stonewall Energy Ctr / Fountain 138 KV	155.2	218	75.9
Gainesville / Michigan Wind 2 138 KV L	Fountain / Stonewall Energy Center 138 KV L	151.0	241	75.9
Vmea Peaking Gen / Woods Dp 138 KV	West Staunton / Woods Dp 138 KV	132.1	234	5.8
Stonewall Energy Center / Blaine 138 KV	Gainesville / Michigan Wind 2 138 KV	124.3	230	70.2
W Staunton / Dean Drive Area Electric Generators 138 KV Line	Stonewall Energy Ctr / Fountain 138 KV	123.3	310	39.3
Novec Paradise / Dean Drive Area Electric Generators 138 KV line	Stonewall Energy Center / Fountain 138 KV	123.3	310	39.3
Blaine / Chaffee Gas Recovery 138 KV	Michigan Wind 2 Substation / Gainesville 138 KV	122.0	264	70.2
Prince William Dp Substation / Clover Hill 138 KV	West Staunton / Woods Dp 138 KV	121.5	232	7.5
Elk Horn / Clover Hill 138 KV	W Staunton / Woods Dp 138 KV	121.5	232	7.5
Stonewall Energy Center / Fountain 138 KV L	Gainesville / Michigan Wind 2 138 KV	118.0	365	70.2
Hoover / Elk Horn 138 KV	W Staunton / Woods Dp 138 KV Line	116.2	227	7.5
Hoover / Menan 138 KV	West Staunton / Woods Dp 138 KV L	115.4	229	7.5
W Staunton / Woods Dp 138 KV	Stonewall Energy Ctr / Fountain 138 KV	114.1	438	30.1
Menan / Norfolk 138 KV	W Staunton / Woods Dp 138 KV Line	113.1	234	7.5
Battery Hts Dp / Toa Vaca 138 KV	Battery Hts Dp / Toa Vaca 138 KV	107.5	446	13.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Battery Heights Dp / Toa Vaca 138 KV Ckt 2	Battery Hts Dp / Toa Vaca 138 KV #1	107.5	446	13.6
Battery Hts Dp Substation / Norfolk 138 KV Ckt 1	Norfolk / Battery Hts Dp 138 KV Line 2	107.5	472	14.3
Norfolk / Battery Hts Dp 138 KV Cir. 2	Battery Heights Dp / Norfolk Substation 138 KV #1	107.5	472	14.3
Colonial / Idaho Springs 138 KV	Colonial / Sideburn 138 KV Line	106.3	492	17.3
Sideburn / Colonial 138 KV	Colonial / Idaho Springs 138 KV	105.6	416	13.5
Colonial / Clifton 138 KV Line 1	Colonial / Clifton 138 KV Ckt 2	105.4	537	17.2
Clifton / Colonial 138 KV Line 2	Colonial / Clifton 138 KV Ckt 1	105.4	537	17.2
Norfolk / Clifton 138 KV line	Clifton / Norfolk 138 KV Circuit 2	105.0	562	17.4
Clifton Substation / Norfolk 138 KV Ckt 2	Clifton / Norfolk 138 KV #1	105.0	562	17.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gainesville / Novec Paradise 138 KV	Rebuild line with higher-capacity conductor	\$1,469,000	48% (\$702,000)
Gainesville / Michigan Wind 2 138 KV L	Rebuild line with higher-capacity conductor	\$2,021,000	48% (\$966,000)
Vmea Peaking Gen / Woods Dp 138 KV	Rebuild line with higher-capacity conductor	\$7,123,000	22% (\$1,551,000)
Stonewall Energy Center / Blaine 138 KV	Rebuild line with higher-capacity conductor	\$1,849,000	77% (\$1,415,000)
W Staunton / Dean Drive Area Electric Generators 138 KV Line	Rebuild line with higher-capacity conductor	\$754,000	47% (\$356,000)
Novec Paradise / Dean Drive Area Electric Generators 138 KV line	Rebuild line with higher-capacity conductor	\$2,828,000	47% (\$1,333,000)
Blaine / Chaffee Gas Recovery 138 KV	Rebuild line with higher-capacity conductor	\$1,637,000	77% (\$1,253,000)
Prince William Dp Substation / Clover Hill 138 KV	Rebuild line with higher-capacity conductor	\$2,279,000	32% (\$726,000)
Elk Horn / Clover Hill 138 KV	Rebuild line with higher-capacity conductor	\$1,148,000	32% (\$366,000)
Stonewall Energy Center / Fountain 138 KV L	Rebuild line with higher-capacity conductor	\$1,411,000	42% (\$598,000)
Elk Horn Substation / Hoover 138 KV L	Rebuild line with higher-capacity conductor	\$1,793,000	32% (\$571,000)
Hoover / Menan 138 KV	Rebuild line with higher-capacity conductor	\$1,516,000	32% (\$483,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wood Dp / W Staunton 138 KV line	Rebuild line with higher-capacity conductor	\$804,000	31% (\$252,000)
Norfolk / Menan 138 KV	Rebuild line with higher-capacity conductor	\$4,641,000	32% (\$1,479,000)
Battery Hts Dp / Toa Vaca 138 KV	Reconductor line	\$361,000	20% (\$71,000)
Battery Heights Dp / Toa Vaca 138 KV Ckt 2	Reconductor line	\$388,000	20% (\$76,000)
Battery Hts Dp Substation / Norfolk 138 KV Ckt 1	Reconductor line	\$268,000	20% (\$53,000)
Norfolk / Battery Hts Dp 138 KV Cir. 2	Reconductor line	\$421,000	20% (\$83,000)
Colonial / Idaho Springs 138 KV	Reconductor line	\$612,000	16% (\$98,000)
Colonial / Sideburn 138 KV	Reconductor line	\$2,192,000	19% (\$424,000)
Clifton Substation / Colonial 138 KV Cir. 1	Reconductor line	\$835,000	16% (\$135,000)
Clifton / Colonial 138 KV Cir. 2	Reconductor line	\$695,000	16% (\$113,000)
Norfolk / Clifton 138 KV Circuit 1	Reconductor line	\$1,023,000	16% (\$166,000)
Clifton / Norfolk 138 KV Ckt 2	Reconductor line	\$1,055,000	16% (\$171,000)
	Total allocated to project		\$13,441,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.